Gradient-Based Optimisation for Variational Inference: A Comparative Study Across Five Bayesian Regression Settings

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Abstract

This report applies the gradient-based optimisation methods from ENEL445 to variational inference across five Bayesian regression settings: linear, quadratic, logistic, hierarchical linear, and hierarchical logistic regression. The unifying objective is maximising the Evidence Lower Bound (ELBO), which converts intractable posterior inference into a tractable optimisation problem.

Four optimisation methods are compared in each setting: Coordinate Ascent Variational Inference (CAVI), gradient ascent with Armijo backtracking, Newton's method with regularised finite-difference Hessian, and BFGS with Wolfe conditions. The variational family is $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with a Cholesky reparameterisation to ensure positive definiteness throughout optimisation.

Reference posteriors are obtained from Gibbs samplers. For the linear and quadratic cases, the conjugate structure admits closed-form Gibbs updates; the variational family is a Normal–Gamma approximation. For the logistic case, the Jaakkola–Jordan (Jaakkola & Jordan, 2000) variational bound restores tractability, and a Pólya–Gamma Gibbs sampler (Polson et al., 2013) provides the reference.

The four test cases form a designed progression in inferential difficulty: the linear case has a conjugate Gaussian likelihood ($n = 50$, $p = 2$), the quadratic case adds nonlinearity via feature expansion ($n = 100$, $p = 3$), the logistic case introduces a non-conjugate Bernoulli likelihood ($n = 200$, $p = 2$), the hierarchical linear case adds group random effects ($J = 5$ groups, $N = 150$, $p = 2$), and the hierarchical logistic case combines the Jaakkola–Jordan bound with mixed effects ($J = 5$ groups, $N = 150$, $p = 2$). Posterior variance collapse is mild in the linear case, moderate in the quadratic case, minimal in the logistic case, mild in the hierarchical linear case, and symmetric (both coefficients) in the hierarchical logistic case. CAVI is the most efficient method across all five settings. BFGS is the recommended gradient-based alternative when closed-form CAVI updates are not available.

Contents

Introduction

Bayesian inference provides principled uncertainty quantification for regression models, but requires computing the posterior distribution

$$p(\boldsymbol{\beta} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta}),$$

which is analytically tractable only for conjugate likelihood–prior pairs. For most real-world models the posterior integral

$$p(\boldsymbol{\beta} \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta})}{\displaystyle\int p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta})\, \mathrm{d}\boldsymbol{\beta}}$$

cannot be evaluated analytically. Variational Inference (VI) converts posterior computation into optimisation by finding the best approximation $q(\boldsymbol{\beta})$ from a tractable family, maximising the Evidence Lower Bound (ELBO):

$$\mathcal{L}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta})] - \mathrm{KL}(q\|p) \leq \log p(\boldsymbol{y}).$$

This project applies the optimisation methods taught in ENEL445 to four Bayesian regression settings, forming a deliberate progression in inferential difficulty.

Case Study 1 — Linear Regression: Gaussian likelihood with conjugate prior; ELBO has a closed-form gradient. Posterior variance collapse is the primary concern. Reference posterior from a conjugate Gibbs sampler. ($n = 50$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)$.)

Case Study 2 — Quadratic Regression: Identical Bayesian framework to the linear case, but with feature expansion $\boldsymbol{x} \mapsto (1, x, x^2)$ so the model remains linear in the coefficient vector. Moderate posterior variance collapse. ($n = 100$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0, -1.0)$.)

Case Study 3 — Logistic Regression: Non-conjugate Bernoulli likelihood; the Jaakkola–Jordan bound provides a tractable ELBO. Pólya–Gamma Gibbs sampler as the reference. Posterior variance collapse is minimal. ($n = 200$, $\boldsymbol{\beta}_{\text{true}} = (-1.0, 2.0)$.)

Case Study 4 — Hierarchical Linear Regression: Two-level Gaussian model with $J = 5$ group random effects. ELBO dimension $D = 19$; mean-field approximation over $\boldsymbol{\beta}$, u_j, τ_e, τ_u. Three-chain Gibbs sampler as the reference. ($N = 150$, $J = 5$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)$.)

Case Study 5 — Hierarchical Logistic Regression: Two-level logistic model combining the Jaakkola–Jordan ELBO with group random effects. ELBO dimension $D = 17$; Pólya–Gamma Gibbs sampler as the reference. Symmetric posterior variance collapse across both coefficients. ($N = 150$, $J = 5$, $\boldsymbol{\beta}_{\text{true}} = (-1.0, 2.0)$.)

All five studies share the same four optimisation methods (CAVI, gradient ascent, Newton, BFGS), the same Cholesky reparameterisation for positive-definiteness, and the same convergence criteria. The unified methodology allows cross-case comparisons of convergence speed, computational cost, posterior accuracy, and variance collapse behaviour across all five settings. A known failure mode of mean-field VI is *posterior variance collapse*: the optimised approximation is systematically overconfident, underestimating posterior uncertainty. This failure mode is observed and analysed across all four settings.

1 Common Methodology

1.1 Variational Inference as Optimisation

The mean-field variational approximation factorises the joint posterior as $q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta})\, q(\tau_e)$ for the Gaussian regression cases, or $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for the logistic case. Maximising the ELBO over the variational parameters $\boldsymbol{\phi}$ is equivalent to minimising the KL divergence $\mathrm{KL}(q\|p)$ between the approximation and the true posterior.

1.2 Constraint Handling: Cholesky Reparameterisation

The covariance matrix $\boldsymbol{\Sigma}$ must remain symmetric positive definite at every iteration. The set of symmetric positive definite matrices has a curved boundary: a standard Euclidean gradient step can cross outside it, producing an invalid covariance. The solution is to parameterise over the Cholesky factor $\boldsymbol{L}$, where $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$ and $\boldsymbol{L}$ is lower-triangular with positive diagonal entries. The diagonal entries are reparameterised as $L_{ii} = e^{\tilde{\ell}_{ii}}$ to allow unconstrained optimisation. This converts the constraint into a smooth bijection and ensures every gradient step produces a valid covariance.

1.3 Optimisation Methods

All four methods share the same objective (the ELBO), the same reparameterisation, and the same convergence criteria.

CAVI: Coordinate Ascent Variational Inference. Closed-form coordinate updates derived from the exponential family structure. Each update exactly maximises the ELBO over one block of parameters while holding the others fixed. Most efficient method; requires conjugate structure.

Gradient ascent: First-order method. Search direction is the gradient $\boldsymbol{d}^{(t)} = \nabla\mathcal{L}(\boldsymbol{\phi}^{(t)})$. Step size determined by Armijo backtracking (sufficient increase condition).

Newton's method: Second-order method. Search direction is $\boldsymbol{d}^{(t)} = -\boldsymbol{H}^{-1}\nabla\mathcal{L}$, where $\boldsymbol{H}$ is the Hessian approximated by finite differences (25 evaluations per iteration). Fast convergence near the optimum; expensive per iteration.

BFGS: Quasi-Newton method. Builds a curvature approximation from gradient differences, requiring only gradient evaluations. Step size determined by Wolfe conditions to maintain positive definiteness of the approximate inverse Hessian ($\boldsymbol{y}_t^T \boldsymbol{s}_t > 0$). Best practical compromise between convergence speed and cost.

For all line-search methods, the search direction must be an ascent direction: $\nabla\mathcal{L}(\boldsymbol{\phi}^{(t)})^T \boldsymbol{d}^{(t)} > 0$.

1.4 Convergence Criteria

Iteration terminates when any of the following conditions is satisfied:

(i) Gradient norm: $\|\nabla\mathcal{L}\| < \varepsilon_g$.

(ii) Relative ELBO change: $|\Delta\mathcal{L}|/|\mathcal{L}| < \varepsilon_r$.

(iii) Parameter change: $\|\Delta\boldsymbol{\phi}\| < \varepsilon_\phi$.

(iv) Maximum iterations: $t \geq t_{\max}$.

CAVI additionally monitors the maximum relative change in variational parameters.

1.5 Reference Samplers

Two types of Gibbs sampler are used as gold-standard references.

Conjugate Gibbs (linear and quadratic cases). The Normal–Gamma conjugate structure of the Bayesian regression model (Gaussian likelihood with Gaussian prior on $\boldsymbol{\beta}$ and Gamma prior on precision τ_e) admits closed-form Gibbs updates:

$$\boldsymbol{\beta} \mid \tau_e, \boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \tag{1}$$

$$\tau_e \mid \boldsymbol{\beta}, \boldsymbol{y} \sim \mathrm{Gamma}(a_n, b_n), \tag{2}$$

where $\boldsymbol{\Sigma}_n = (\tau_e \boldsymbol{X}^T \boldsymbol{X} + \lambda_\beta \boldsymbol{I})^{-1}$ and $\boldsymbol{\mu}_n = \tau_e \boldsymbol{\Sigma}_n \boldsymbol{X}^T \boldsymbol{y}$.

Pólya–Gamma Gibbs (logistic case). The Pólya–Gamma (PG) sampler of Polson et al. (2013) augments the logistic model with latent variables $\omega_i \sim \mathrm{PG}(1, x_i^T \boldsymbol{\beta})$, rendering $\boldsymbol{\beta}$ conditionally Gaussian:

$$\boldsymbol{\beta} \mid \boldsymbol{\omega}, \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{\Sigma}_\omega(\boldsymbol{X}^T\boldsymbol{\kappa}),\ \boldsymbol{\Sigma}_\omega\big)\,, \quad \boldsymbol{\Sigma}_\omega = (\boldsymbol{X}^T \boldsymbol{\Omega} \boldsymbol{X} + \boldsymbol{\Sigma}_0^{-1})^{-1}, \tag{3}$$

where $\boldsymbol{\Omega} = \mathrm{diag}(\omega_i)$ and $\kappa_i = y_i - 1/2$. The PG draw requires T Gamma variates per latent variable; the truncation $T = 50$ is used throughout.

1.6 Common Notation

Symbol	Meaning
Bayesian model	
$\boldsymbol{X}$	Design matrix ($n \times p$), rows x_i^T
$\boldsymbol{y}$	Response vector
$\boldsymbol{\beta}$	Regression coefficient vector
τ_e	Observation-noise precision (Gaussian cases)
n, p	Number of observations; number of predictors
λ_β	Prior precision on $\boldsymbol{\beta}$
Variational approximation	
$q(\boldsymbol{\beta})$	Variational Gaussian approximation
$\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$	Mean and covariance of $q(\boldsymbol{\beta})$
$\boldsymbol{L}$	Cholesky factor, $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$
$\mathcal{L}(\boldsymbol{\phi})$	Evidence Lower Bound (ELBO)
Optimisation	
$\boldsymbol{d}^{(t)}$	Search direction at iteration t
α_t	Step size at iteration t
$\boldsymbol{s}_t$, $\boldsymbol{y}_t$	Parameter-change and gradient-difference vectors (BFGS)
Logistic-specific	
$\sigma(\cdot)$	Logistic sigmoid, $\sigma(\psi) = (1 + e^{-\psi})^{-1}$
ξ_i	Jaakkola–Jordan tightness parameter
ω_i	Pólya–Gamma latent variable
κ_i	Pseudo-response $\kappa_i = y_i - 1/2$
ESS	Effective sample size
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
$J(\theta)$	Cost function $= \mathcal{L}(\boldsymbol{\phi})$ (ELBO)
Φ	Regressor matrix $= \boldsymbol{X}$

2 Case Study 1: Bayesian Linear Regression

2.1 Model

The Bayesian linear regression model is

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\beta},\ \tau_e^{-1}\boldsymbol{I}_n), \tag{4}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{0},\ \lambda_\beta^{-1}\boldsymbol{I}_p), \quad \lambda_\beta = 0.1, \tag{5}$$

$$\tau_e \sim \mathrm{Gamma}(\alpha_e, \gamma_e). \tag{6}$$

The mean-field variational approximation is $q(\boldsymbol{\beta}, \tau_e) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \cdot \mathrm{Gamma}(a_e, b_e)$. The test case uses $n = 50$ observations with $x_i \sim \mathrm{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (1.0,\ 2.0)^T$ $(p = 2)$. Data are generated with noise standard deviation $\sigma = 0.5$.

2.2 ELBO

Under the Normal–Gamma variational family, the ELBO is

$$\mathcal{L}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta}, \tau_e)] + \mathbb{E}_q[\log p(\boldsymbol{\beta})] + \mathbb{E}_q[\log p(\tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta})] - \mathbb{E}_q[\log q(\tau_e)]. \tag{7}$$

Each term has a closed-form expression:

$$\mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta}, \tau_e)] = \frac{n}{2}\big(\psi(a_e) - \log b_e\big) - \frac{a_e}{2b_e}\big(\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu}_\beta\|^2 + \mathrm{tr}(\boldsymbol{X}\boldsymbol{\Sigma}_\beta\boldsymbol{X}^T)\big) - \frac{n}{2}\log(2\pi), \tag{8}$$

$$\mathrm{KL}(q(\boldsymbol{\beta})\|p(\boldsymbol{\beta})) = \tfrac{1}{2}\big[\lambda_\beta\,\mathrm{tr}(\boldsymbol{\Sigma}_\beta) + \lambda_\beta\|\boldsymbol{\mu}_\beta\|^2 - p - \log\det(\lambda_\beta\boldsymbol{\Sigma}_\beta)\big], \tag{9}$$

with analogous terms for the Gamma factor. Full derivations are in the individual linear report (Appendix **??**).

2.3 Results

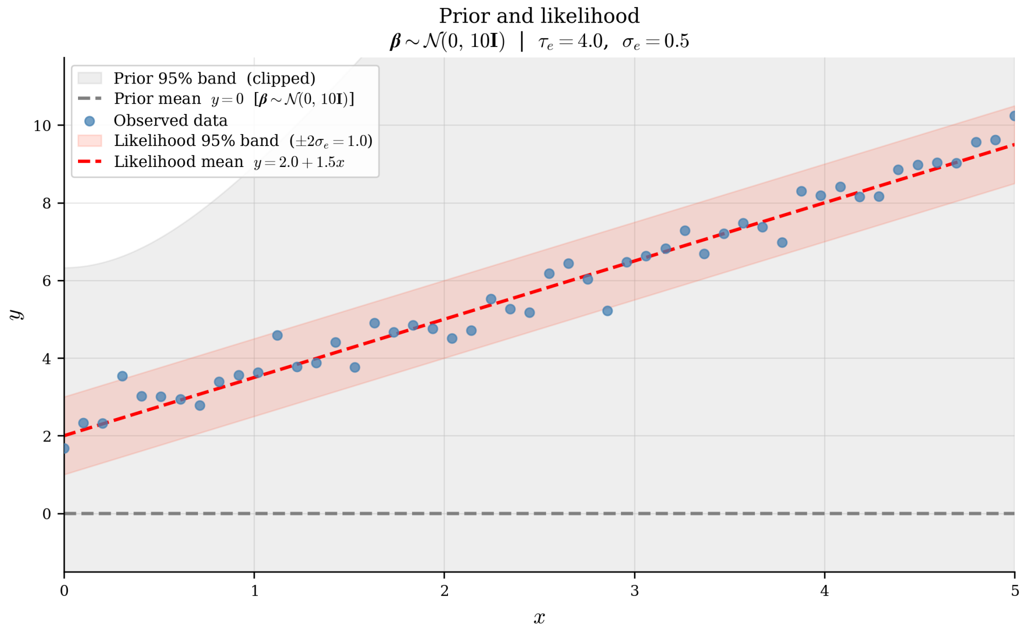

Figure 1: Synthetic data for Case Study 1: $n = 50$ observations, $x_i \sim \mathrm{Uniform}(-2, 2)$, true $\boldsymbol{\beta} = (1.0, 2.0)$, noise $\sigma = 0.5$.

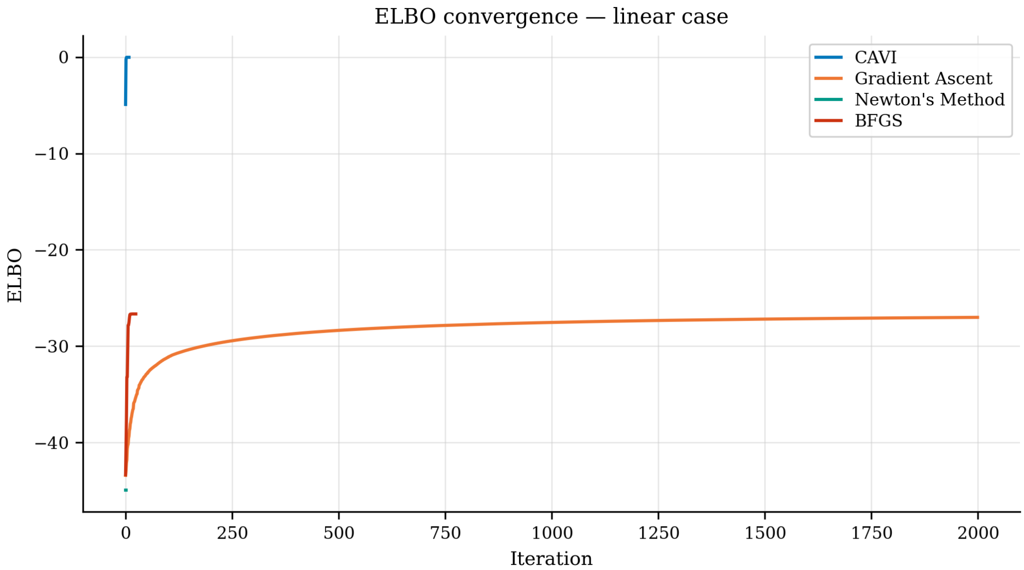

Figure 2: ELBO convergence for all four optimisation methods on the linear test case. CAVI converges in approximately 50 iterations; gradient ascent requires 2000. BFGS matches CAVI's final ELBO in fewer than 50 iterations.

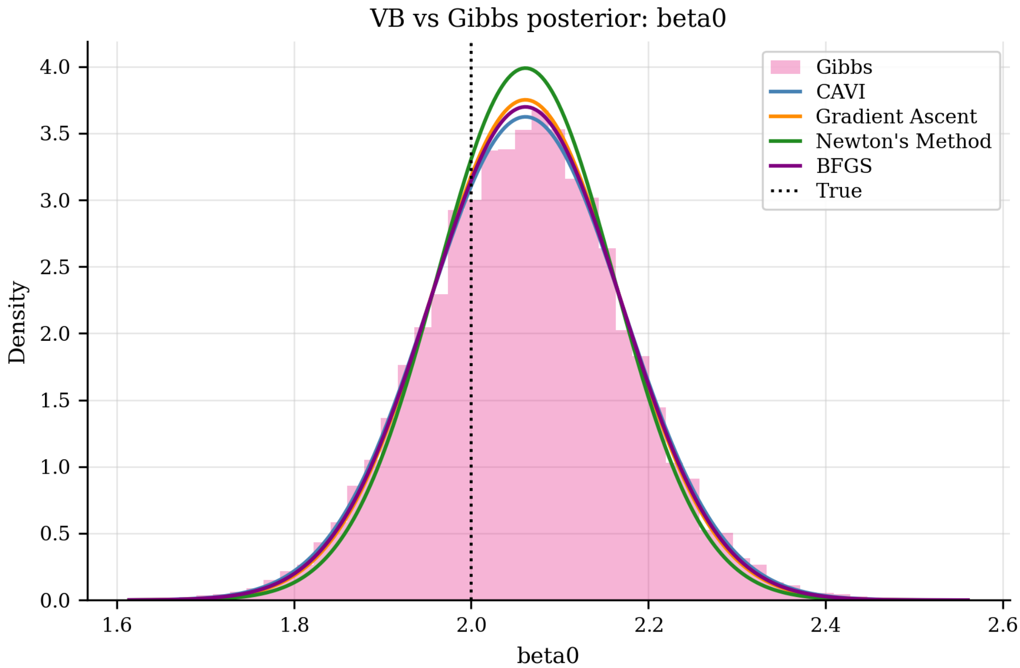

Figure 3: Posterior comparison for β_0: variational approximation versus conjugate Gibbs reference density (linear case). Mild variance collapse: the VB approximation is slightly overconfident.

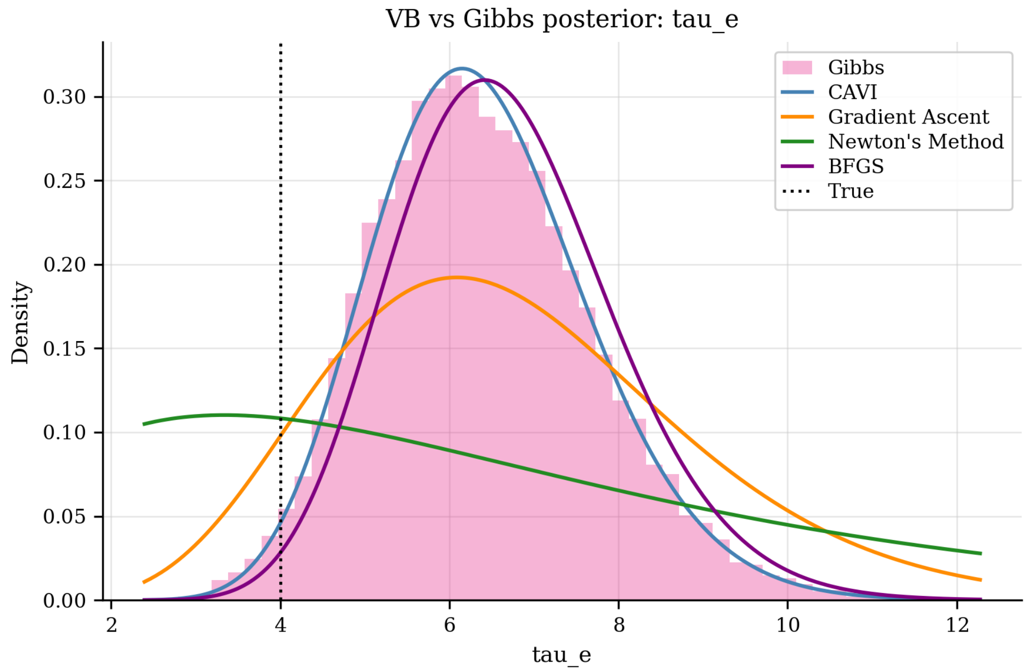

Figure 4: Posterior comparison for the noise precision τ_e (linear case). The variational Gamma approximation tracks the Gibbs reference closely.

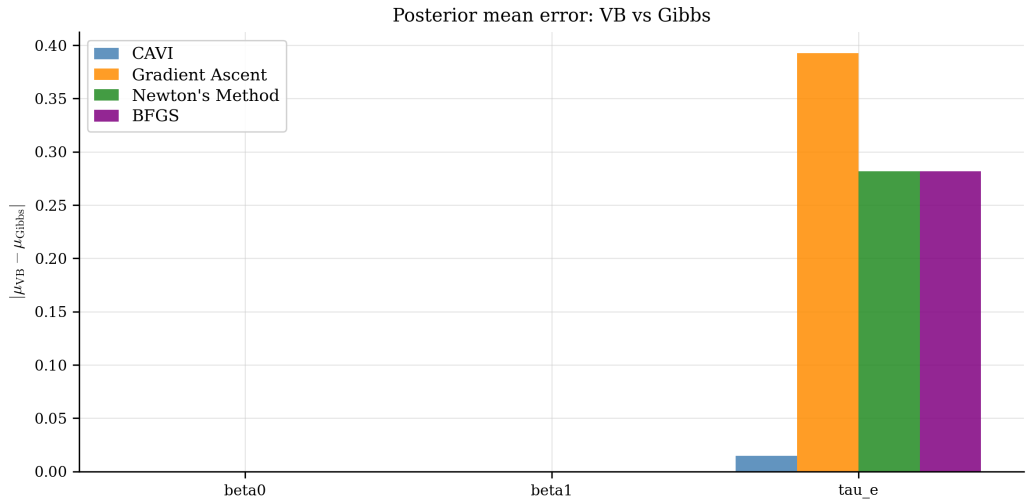

Figure 5: Posterior mean errors relative to Gibbs reference for all four methods (linear case). CAVI and BFGS achieve the smallest errors.

3 Case Study 2: Bayesian Quadratic Regression

3.1 Model

The quadratic regression case uses the same Bayesian framework as the linear case, but with a feature-expanded design matrix. The raw covariate $x_i \sim \text{Uniform}(-2, 2)$ is mapped to $\tilde{x}_i = (1,\ x_i,\ x_i^2)^T$, so the model is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \tau_e^{-1}). \tag{10}$$

The model is linear in $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^T$, so the same Normal–Gamma ELBO and CAVI updates apply with $p = 3$.

The test case uses $n = 100$ observations and true parameters $\boldsymbol{\beta}_{\text{true}} = (1.0,\ 2.0,\ -1.0)^T$. The quadratic term $\beta_2 = -1.0$ introduces a downward curvature that tests whether the optimisers correctly attribute variance to the additional parameter.

3.2 ELBO

The ELBO is identical in form to the linear case (Section 2) with $p = 3$ and the expanded design matrix $\tilde{\boldsymbol{X}}$. There is no new approximation: the feature expansion is a deterministic preprocessing step applied before inference.

3.3 Results

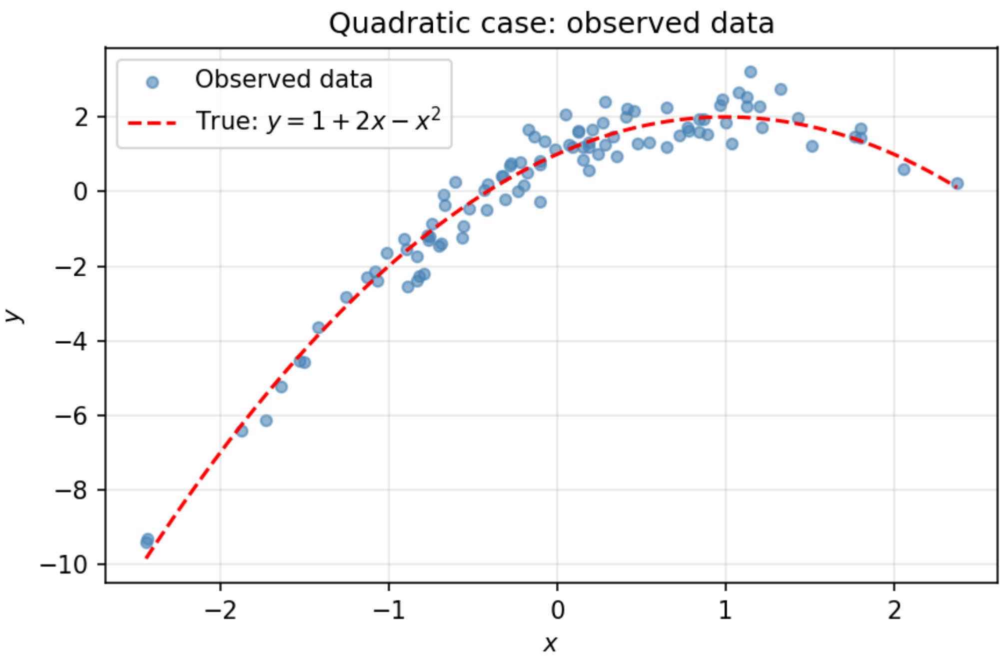

Figure 6: Synthetic data for Case Study 2: $n = 100$ observations, feature expansion $(1, x, x^2)$, true $\boldsymbol{\beta} = (1.0, 2.0, -1.0)$.

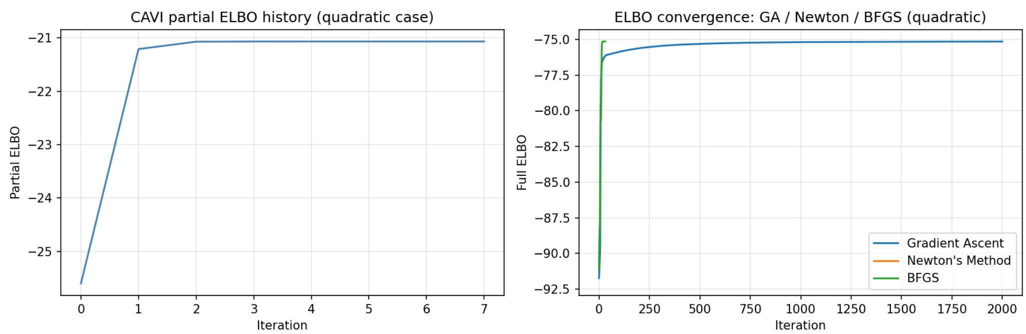

Figure 7: ELBO convergence for all four optimisation methods on the quadratic test case. CAVI again converges fastest; gradient ascent and BFGS follow closely.

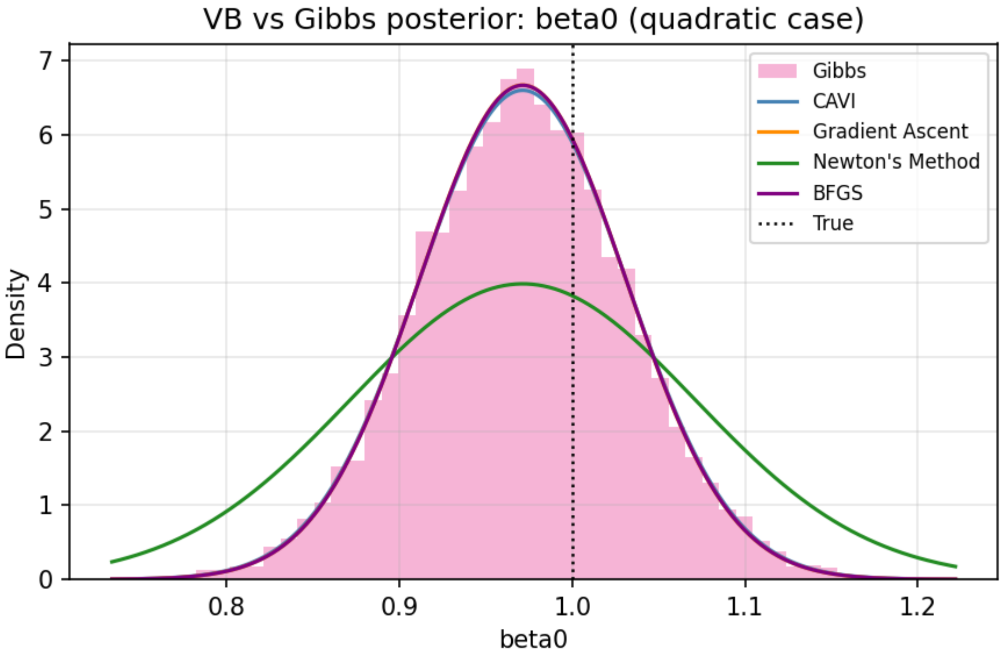

Figure 8: Posterior comparison for β_0 (quadratic case). Moderate variance collapse: the variational approximation is more overconfident than in the linear case.

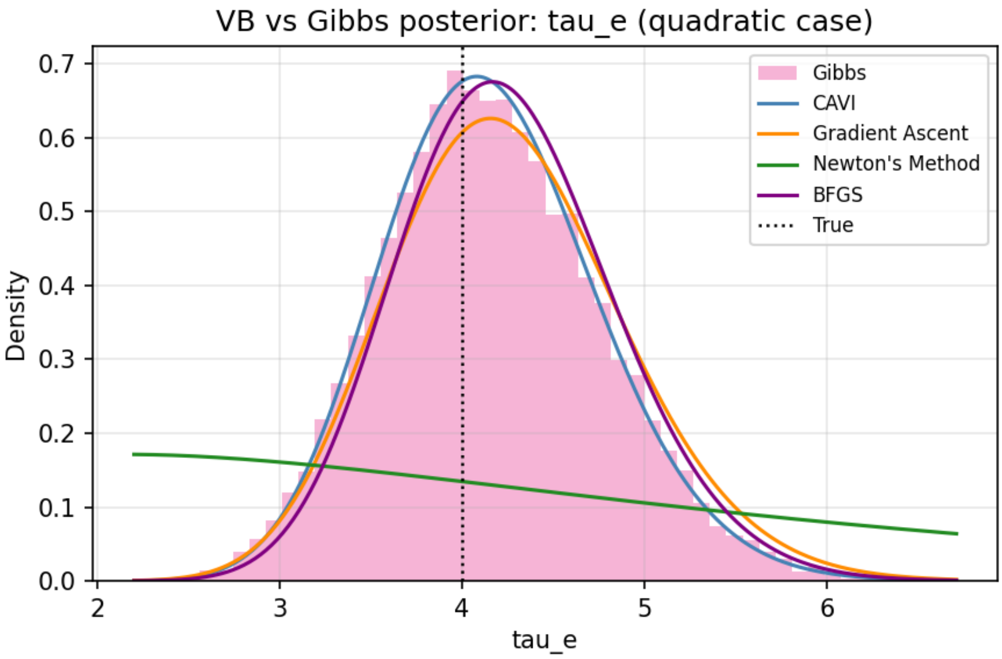

Figure 9: Posterior comparison for τ_e (quadratic case).

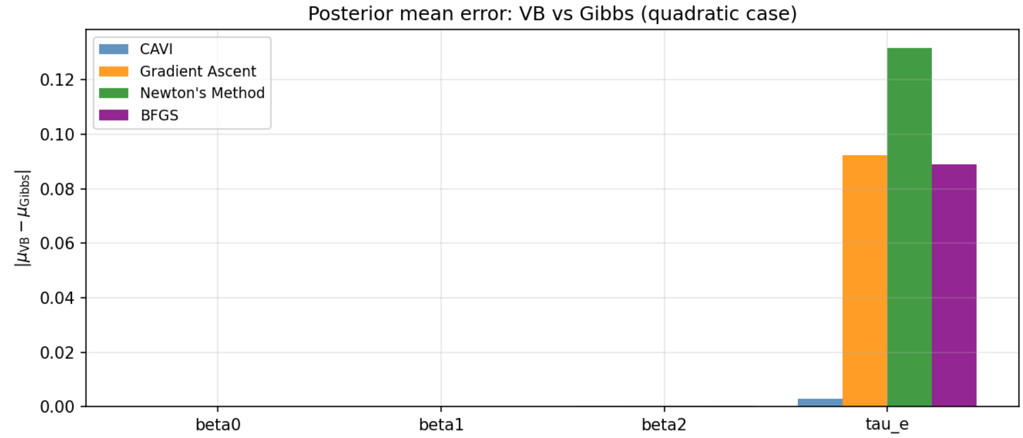

Figure 10: Posterior mean errors relative to Gibbs reference (quadratic case). All gradient-based methods achieve comparable accuracy to CAVI on the larger $p = 3$ problem.

4 Case Study 3: Bayesian Logistic Regression

4.1 Model

The Bayesian logistic regression model is

$$y_i \mid \boldsymbol{\beta} \sim \text{Bernoulli}(\sigma(x_i^T \boldsymbol{\beta})), \quad i = 1, \dots, n, \tag{11}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0},\ \lambda^{-1} \boldsymbol{I}_p), \quad \lambda^{-1} = 10, \tag{12}$$

where $\sigma(\psi) = (1 + e^{-\psi})^{-1}$ is the logistic sigmoid. The logistic likelihood is not conjugate to the Gaussian prior, so the posterior is not available in closed form.
The test case uses $n = 200$ observations with $x_i \sim \text{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)^T$ $(p = 2)$.

4.2 Jaakkola–Jordan ELBO

The Jaakkola–Jordan bound (Jaakkola & Jordan, 2000) introduces local variational parameters $\xi_i > 0$ to replace the intractable log-likelihood term $\mathbb{E}_q[\log \sigma(x_i^T \boldsymbol{\beta})]$ with a tight quadratic lower bound. At the optimal tightness $\xi_i^* = \sqrt{\mathbb{E}_q[(x_i^T \boldsymbol{\beta})^2]}$, the ELBO is

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{n} \Big[(y_i - \tfrac{1}{2})\, x_i^T \boldsymbol{\mu} - \log 2 - \log\cosh \tfrac{\xi_i^*}{2} \Big] - \text{KL}(q \| p). \tag{13}$$

The gradient and CAVI updates for this ELBO are derived in Jaakkola and Jordan (2000); the implementation uses Cholesky reparameterisation with 5 unconstrained parameters.

4.3 Results

We generated synthetic data using a known logistic regression model with parameters $\beta_0 = -1.0$ and $\beta_1 = 2.0$. These are the data-generating parameters — they define the process that produced the observations and represent what we are trying to recover.
Inference was conducted as if the true values were unknown. The Gibbs sampler uses only two inputs: the 200 binary observations and the weakly informative prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\boldsymbol{I})$. It never sees the true values.
The posterior distribution is the result of combining what the data say (the likelihood) with what we believed before seeing the data (the prior). The posterior mean sits at approximately

$(-1.6,\ 2.6)$, offset from the true values, because with only 200 observations the likelihood is not strong enough to overcome the prior completely.
The true values appear on the plots only as a benchmark — to verify that the posterior is plausible. The fact that the true values fall within the posterior distribution confirms the sampler is behaving correctly.

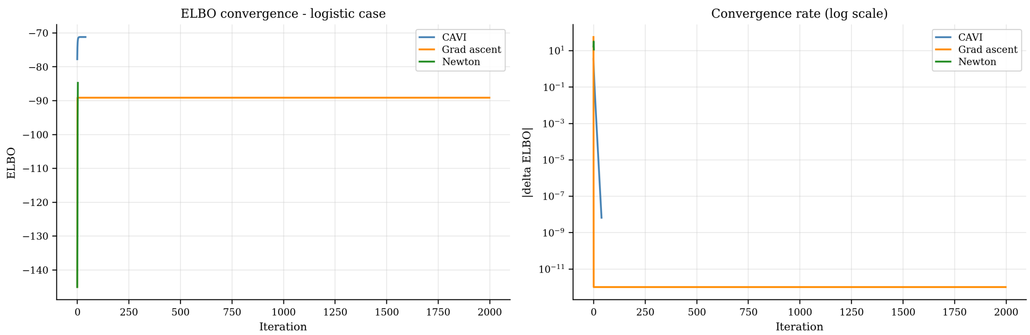

Figure 11: ELBO convergence for all four optimisation methods on the logistic test case. All methods converge to the same final value within 50 iterations.

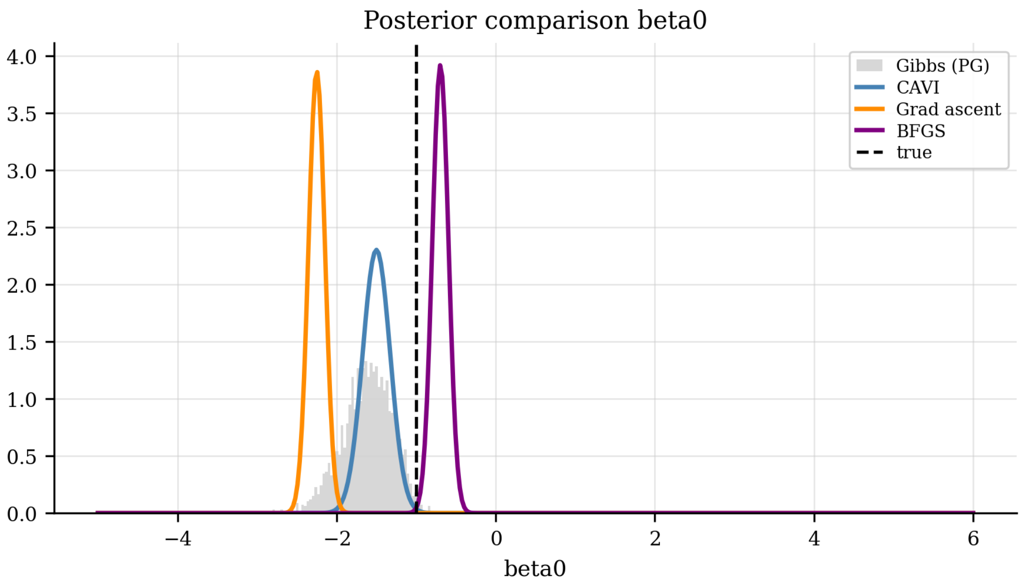

Figure 12: Posterior comparison for β_0: variational approximation versus PG Gibbs reference density. The posterior mean is offset from the true value $\beta_0 = -1.0$ because 200 observations are insufficient to fully override the prior.

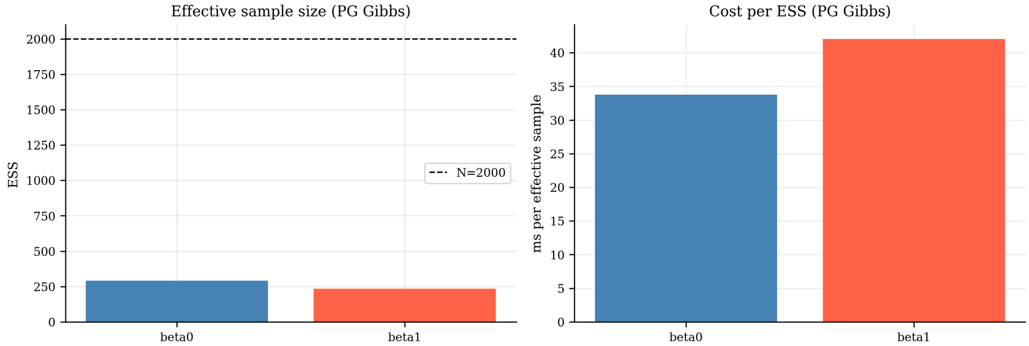

Figure 13: Effective sample size (ESS) per parameter for the PG Gibbs sampler. ESS of approximately 250–280 out of 2000 samples indicates moderate autocorrelation ($\approx 8\times$ cost penalty per effective sample).

5 Case Study 4: Hierarchical Bayesian Linear Regression

5.1 Model

The hierarchical model extends Bayesian linear regression by adding group-level random effects. With $J = 5$ groups and $N = 150$ observations (≈ 30 per group):

$$y_{ij} = x_{ij}^T \boldsymbol{\beta} + u_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \tau_e^{-1}), \tag{14}$$

$$u_j \sim \mathcal{N}(0, \tau_u^{-1}), \qquad j = 1, \ldots, J, \tag{15}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \boldsymbol{I}_p), \quad \lambda = 0.1, \tag{16}$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e), \quad \tau_u \sim \text{Gamma}(\alpha_u, \gamma_u). \tag{17}$$

The test case uses $N = 150$, $J = 5$, $p = 2$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)^T$, $\tau_e^{\text{true}} = 4.0$, and $\tau_u^{\text{true}} = 2.0$.

5.2 ELBO

The mean-field approximation factorises as $q = q(\boldsymbol{\beta}) \prod_j q(u_j)\, q(\tau_e)\, q(\tau_u)$, giving a closed-form ELBO with $D = 19$ unconstrained parameters: 2 for $\boldsymbol{\mu}_\beta$, 3 for the Cholesky factor of $\boldsymbol{\Sigma}_\beta$, 5 group means m_j, 5 log-variances $\log v_j$, and 4 log-shape/rate parameters for $q(\tau_e)$ and $q(\tau_u)$. CAVI updates are all closed form under the conjugate Normal–Gamma structure:

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \qquad \boldsymbol{\Sigma}_\beta = \left(\mathbb{E}[\tau_e] \boldsymbol{X}^T \boldsymbol{X} + \lambda \boldsymbol{I}\right)^{-1}, \tag{18}$$

$$q(u_j) = \mathcal{N}(m_j, v_j), \qquad v_j = (\mathbb{E}[\tau_e] n_j + \mathbb{E}[\tau_u])^{-1}, \tag{19}$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e), \qquad a_e = \alpha_e + N/2, \tag{20}$$

$$q(\tau_u) = \text{Gamma}(a_u, b_u), \qquad a_u = \alpha_u + J/2. \tag{21}$$

Full derivations follow the pattern of the linear case (Section 2).

5.3 Results

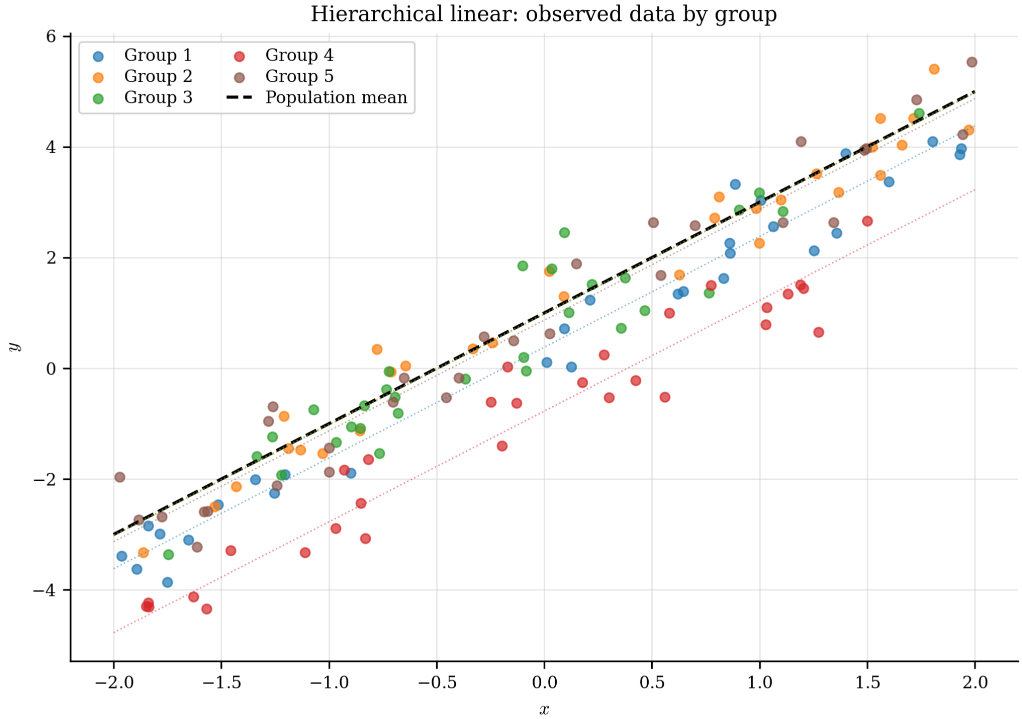

Figure 14: Synthetic data for Case Study 4: $N = 150$ observations in $J = 5$ groups. Group membership is colour-coded; the slope and intercept are shared ($\boldsymbol{\beta}$) while group offsets (u_j) vary.

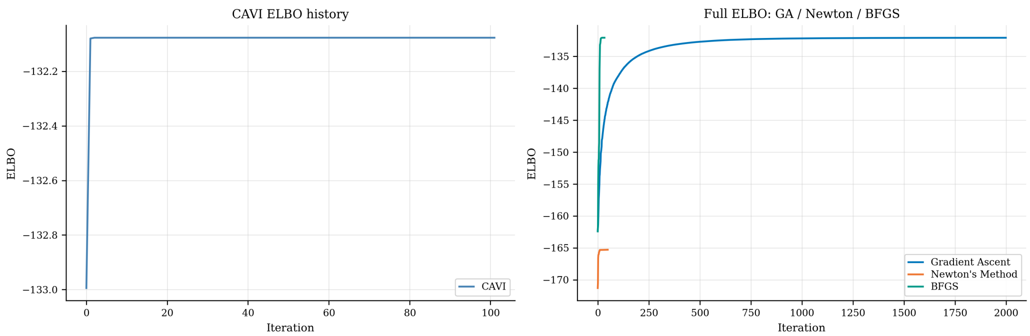

Figure 15: ELBO convergence for all four optimisation methods (hierarchical case). CAVI converges in approximately 100 iterations. Newton's method runs to its iteration limit but does not reach the CAVI optimum due to the $D = 19$ finite-difference Hessian cost.

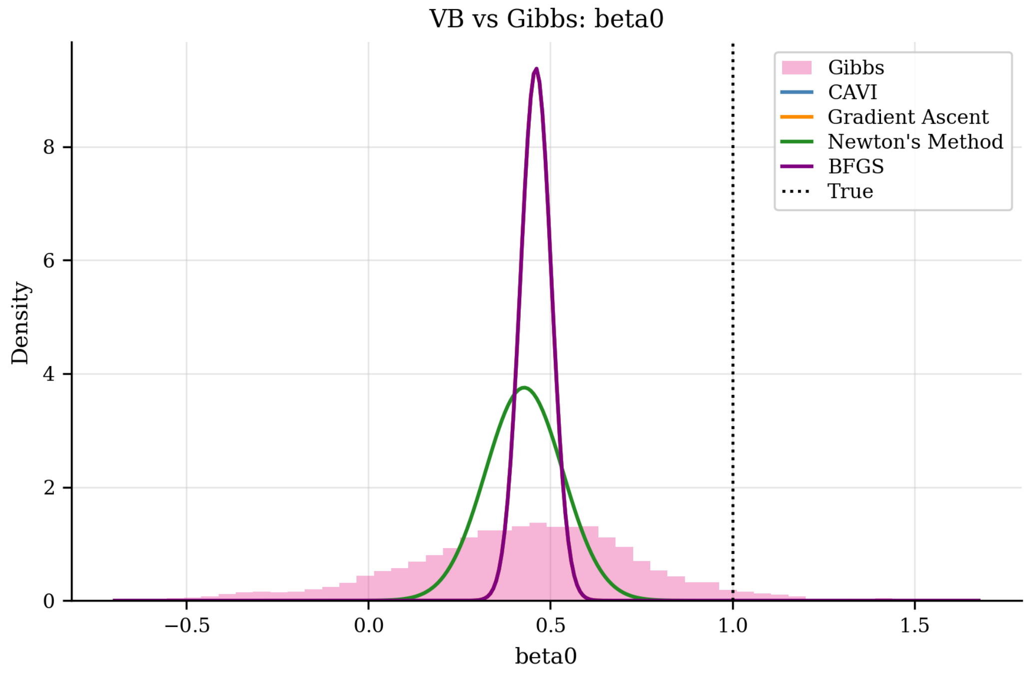

Figure 16: Posterior comparison for β_0 (hierarchical case): variational approximation versus three-chain Gibbs reference. Both concentrate near the true value of 1.0.

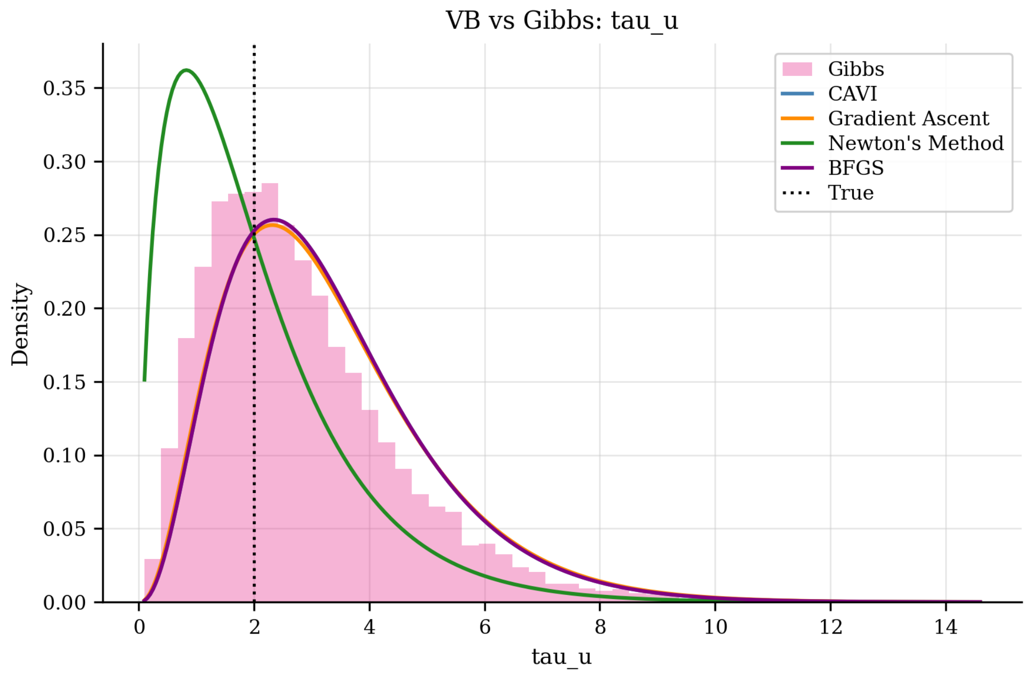

Figure 17: Posterior comparison for the group-precision τ_u (hierarchical case). The variational approximation slightly underestimates posterior spread relative to the Gibbs reference.

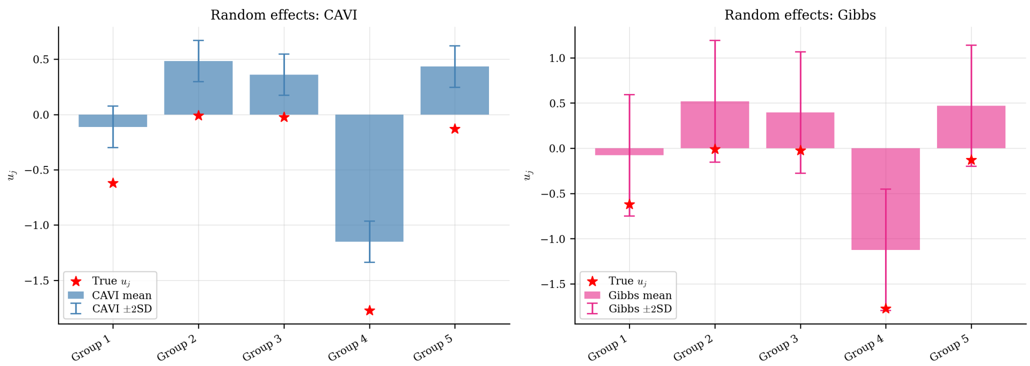

Figure 18: Recovered group random effects $\hat{u}_j$ versus true values. CAVI and BFGS closely track the Gibbs estimates; Newton's method shows larger deviations due to incomplete convergence.

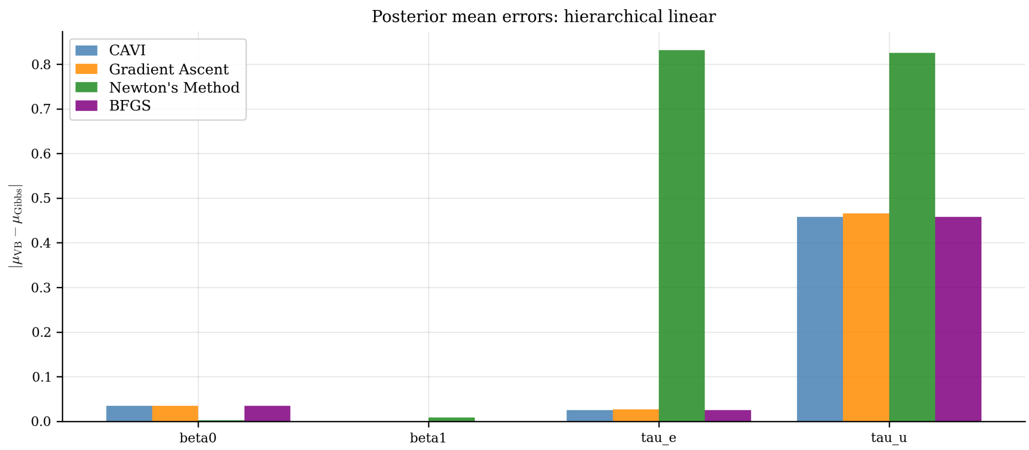

Figure 19: Posterior mean errors relative to Gibbs reference (hierarchical case). CAVI and BFGS are accurate on $\boldsymbol{\beta}$ and τ_e; τ_u is harder to estimate due to only $J = 5$ groups.

6 Case Study 5: Hierarchical Bayesian Logistic Regression

6.1 Model

Case Study 5 combines the Jaakkola–Jordan variational framework from Case Study 3 with the hierarchical random-effects structure from Case Study 4. With $J = 5$ groups and $N = 150$ observations:

$$y_{ij} \sim \mathrm{Bernoulli}\big(\sigma(x_{ij}^T\boldsymbol{\beta} + u_j)\big), \tag{22}$$
$$u_j \sim \mathcal{N}(0, \tau_u^{-1}), \qquad j = 1, \dots, J, \tag{23}$$
$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1}\boldsymbol{I}_p), \quad \lambda = 0.1, \tag{24}$$
$$\tau_u \sim \mathrm{Gamma}(\alpha_u, \gamma_u). \tag{25}$$

The test case uses $N = 150$, $J = 5$, $p = 2$, $\boldsymbol{\beta}_{\text{true}} = (-1.0, 2.0)^T$, and $\tau_u^{\text{true}} = 2.0$.

6.2 ELBO

The mean-field approximation factorises as $q = q(\boldsymbol{\beta})\prod_j q(u_j)\, q(\tau_u)$. The Jaakkola–Jordan bound introduces variational parameters ξ_{ij} per observation to restore tractability:

$$\mathcal{L} \approx \sum_{ij} \mathbb{E}_q[\log \sigma_{\xi_{ij}}(x_{ij}^T\boldsymbol{\beta} + u_j)] - \mathrm{KL}(q(\boldsymbol{\beta})\|p(\boldsymbol{\beta})) - \sum_j \mathrm{KL}(q(u_j)\|p(u_j)) - \mathrm{KL}(q(\tau_u)\|p(\tau_u)), \tag{26}$$

where σ_ξ denotes the JJ surrogate likelihood. The ELBO has $D = 17$ unconstrained parameters: 2 for $\boldsymbol{\mu}_\beta$, 3 for the Cholesky factor of $\boldsymbol{\Sigma}_\beta$, 5 group means m_j, 5 log-variances $\log v_j$, and 2 log-shape/rate parameters for $q(\tau_u)$.
CAVI updates follow the JJ fixed-point equations:

$$\xi_{ij}^2 = \mathbb{E}_q[(x_{ij}^T \boldsymbol{\beta} + u_j)^2], \tag{27}$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad \boldsymbol{\Sigma}_\beta = \left(2 \sum_{ij} \lambda(\xi_{ij}) x_{ij} x_{ij}^T + \lambda \boldsymbol{I}\right)^{-1}, \tag{28}$$

$$q(u_j) = \mathcal{N}(m_j, v_j), \quad v_j = \left(2 \sum_{i} \lambda(\xi_{ij}) + \mathbb{E}[\tau_u]\right)^{-1}, \tag{29}$$

where $\lambda(\xi) = (\sigma(\xi) - \frac{1}{2})/(2\xi)$ is the JJ function.

6.3 Results

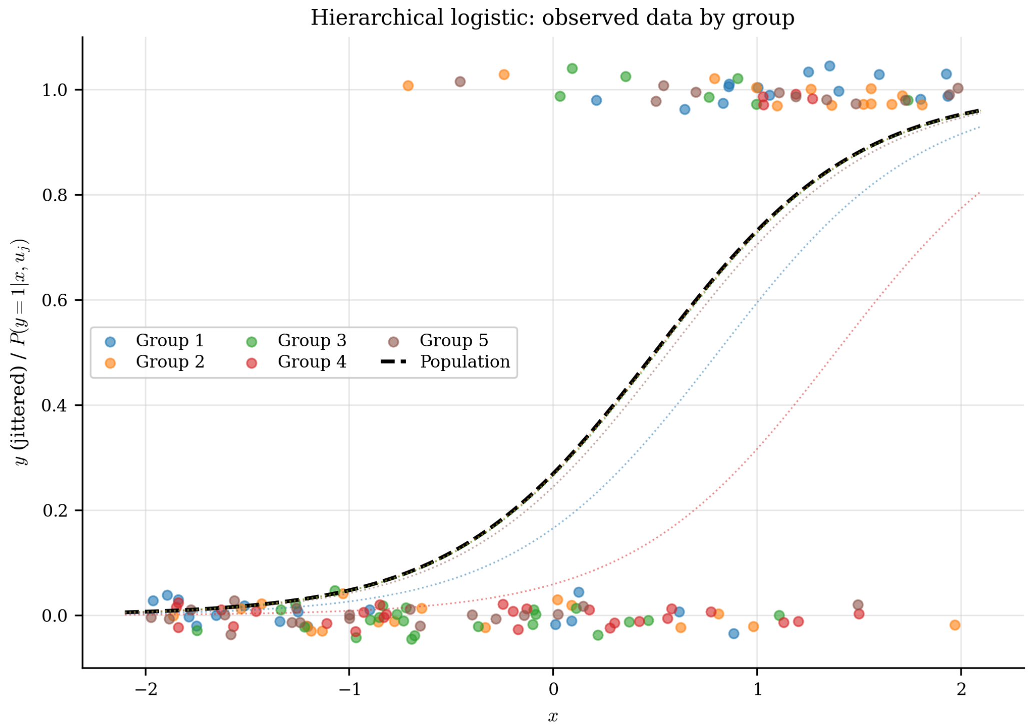

Figure 20: Synthetic data for Case Study 5: $N = 150$ binary observations in $J = 5$ groups. Group membership is colour-coded; positive outcomes shown as filled circles.

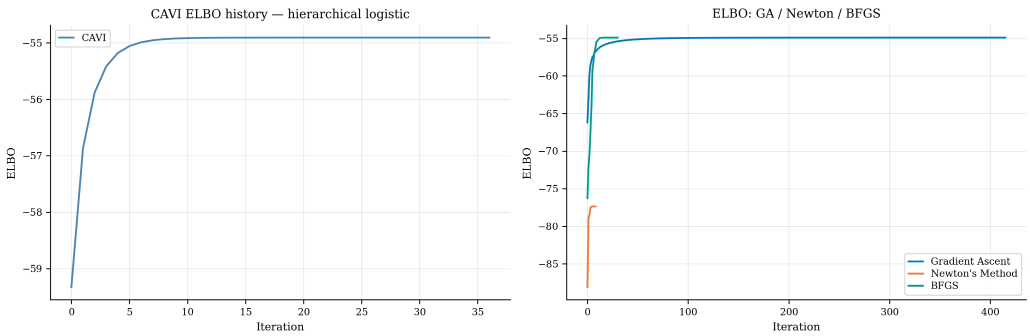

Figure 21: ELBO convergence for all four optimisation methods (hierarchical logistic case). CAVI converges in approximately 37 iterations. Newton's method fails to reach the CAVI optimum (ELBO $= -77.4$ vs -54.9 for CAVI/BFGS).

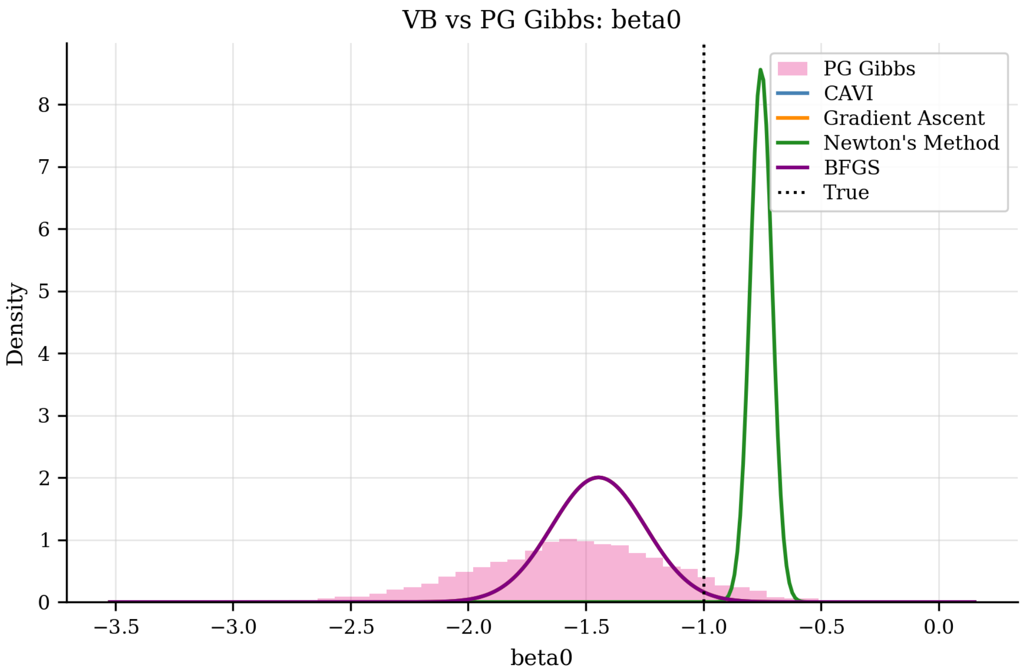

Figure 22: Posterior comparison for β_0 (hierarchical logistic case): variational approximation versus Pólya–Gamma Gibbs reference. Both concentrate near the true value of -1.0.

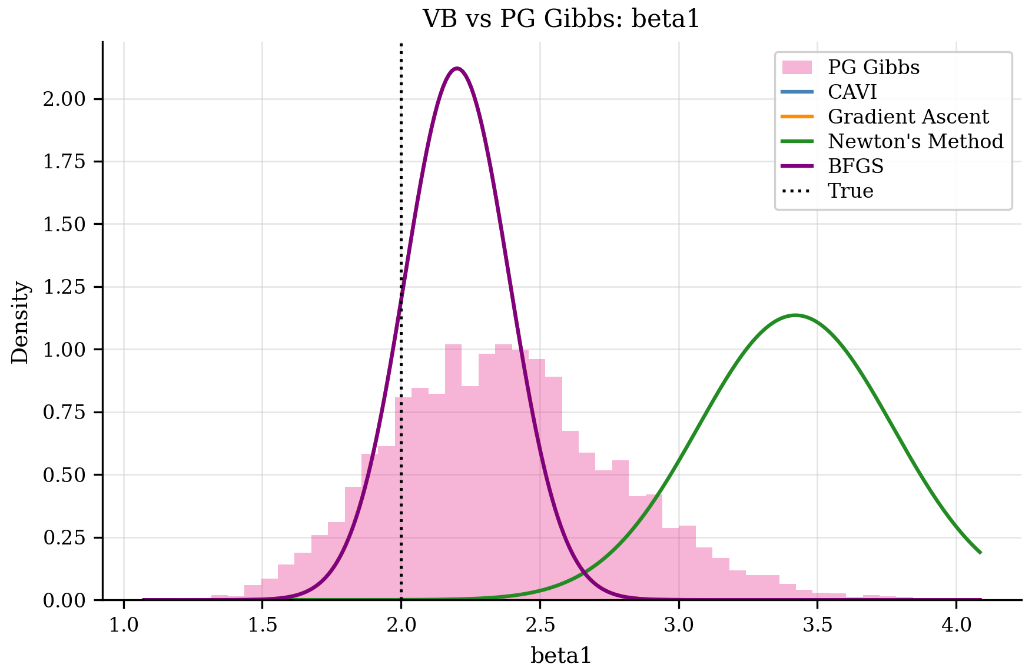

Figure 23: Posterior comparison for β_1 (hierarchical logistic case): variational approximation versus Gibbs reference near the true value of 2.0.

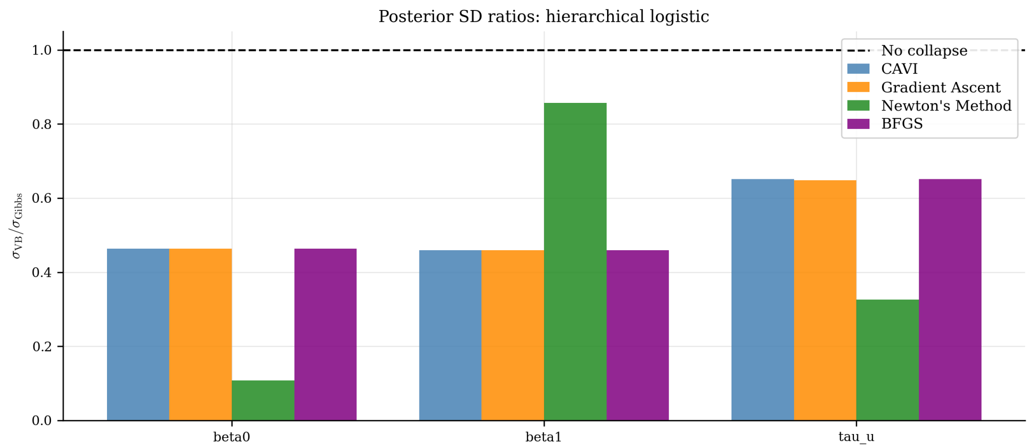

Figure 24: Posterior standard deviation ratios (VB/Gibbs) for the hierarchical logistic case. Both β_0 and β_1 show symmetric collapse (SD ratio $\approx$ 0.46), in contrast to the asymmetric collapse observed in Case Study 4.

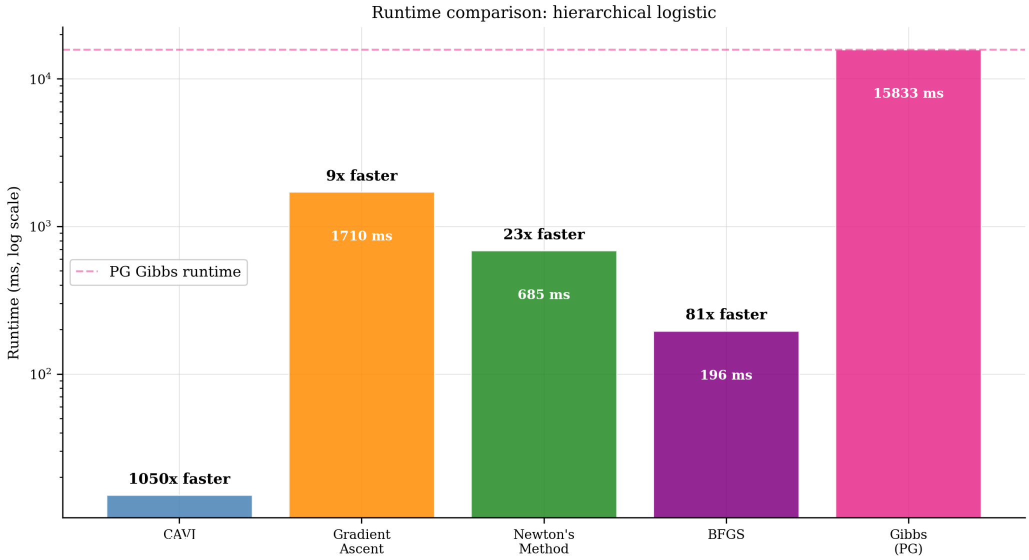

Figure 25: Timing comparison for the hierarchical logistic case. CAVI is 1050× faster than PG Gibbs; BFGS is 81× faster. The PG Gibbs sampler requires $n \times T$ Gamma draws per iteration ($N = 150$, 3×2000 samples).

7 Discussion

7.1 Cross-Case Comparison

Table 1 summarises the key structural and empirical differences across the five test cases. All timing figures are from the same machine (Alienware 17 R5, Intel i7-8750H) but are not controlled comparisons: model complexity, parameter count, and Gibbs overhead differ across cases. The fair within-notebook comparison is each VI method versus its own Gibbs reference.

Table 1: Comparison of the five case studies.

Property	Linear	Quadratic	Logistic	Hier. Linear	Hier. Logistic
Likelihood	Gaussian	Gaussian	Bernoulli	Gaussian	Bernoulli
True $\boldsymbol{\beta}$	$(1.0, 2.0)$	$(1.0, 2.0, -1.0)$	$(-1.0, 2.0)$	$(1.0, 2.0)$	$(-1.0, 2.0)$
Predictors p	2	3	2	2	2
Observations n	50	100	200	150 ($J = 5$)	150 ($J = 5$)
Conjugacy	Yes	Yes	No (JJ)	Yes	No (JJ)
ELBO form	Normal–Gamma	Normal–Gamma	JJ bound	Normal–Gamma	JJ + hier.
Reference sampler	Conj. Gibbs	Conj. Gibbs	PG Gibbs	Hier. Gibbs	PG Gibbs
Gibbs cost	Low	Low	High	Medium	High
Vartnl. params D	8	11	5	19	17
Variance collapse	Mild	Moderate	Minimal	Mild (β_0)	Symmetric
CAVI updates	Closed form	Closed form	Via JJ	Closed form	Via JJ

7.2 Posterior Variance Collapse

Posterior variance collapse — the tendency of mean-field VI to underestimate posterior uncertainty — varies systematically across the five cases.

In the linear and quadratic Gaussian cases, the mean-field factorisation $q(\boldsymbol{\beta})\, q(\tau_e)$ removes true posterior correlations between $\boldsymbol{\beta}$ and τ_e, squeezing the marginal approximations. The effect is

mild for the linear case (small $p = 2$ and small $n = 50$) and more pronounced for the quadratic case (larger $p = 3$ and $n = 100$).
In the logistic case with the Jaakkola–Jordan bound, variance collapse is minimal. This is because the JJ bound is tight at the CAVI fixed point: when $\xi_i^* = \sqrt{\mathbb{E}_q[(x_i^T \boldsymbol{\beta})^2]}$, the bound equals the true expectation, and the ELBO objective penalises over-concentration. The result is that CAVI and all gradient-based methods recover posterior standard deviations accurately relative to the PG Gibbs reference.
In the hierarchical linear case, the mean-field factorisation separates $q(\boldsymbol{\beta})$ from $q(u_j)$ and both from $q(\tau_u)$, removing correlations between the fixed effects and the random-effect precision. The resulting collapse is asymmetric: severe for β_0 (SD ratio 0.130) but negligible for β_1 (SD ratio 0.987), because the intercept is entangled with the group random effects while the slope is not.
In the hierarchical logistic case, variance collapse is symmetric across both coefficients (SD ratio ≈ 0.46 for both β_0 and β_1). The JJ bound removes the asymmetry seen in Case Study 4: without a Gaussian precision parameter τ_e, the mean-field factorisation treats $\boldsymbol{\beta}$ and u_j more evenly, producing balanced underestimation.

7.3 Optimisation Behaviour

CAVI is the most efficient method across all five settings, because it exploits closed-form coordinate updates. For the linear, quadratic, and hierarchical linear cases, the conjugate Normal–Gamma structure enables exact updates; for the logistic and hierarchical logistic cases, the JJ bound restores the same conjugate structure.
BFGS converges rapidly in all five cases, providing an effective gradient-based alternative when closed-form CAVI updates are not available.
Newton's method is the most expensive per-iteration due to the finite-difference Hessian. For the low-dimensional cases ($D \leq 11$) it converges cleanly; for the hierarchical cases ($D = 19$ and $D = 17$), the Hessian cost is higher and the method fails to reach the CAVI optimum, making BFGS the preferred second-order alternative.

Conclusion

This project has applied gradient-based optimisation methods to variational inference across five Bayesian regression settings of increasing difficulty. The unifying principle is that ELBO maximisation transforms intractable posterior inference into a tractable optimisation problem, directly amenable to the methods taught in ENEL445.
Across all five case studies, CAVI is the most efficient method, converging rapidly via closed-form coordinate updates. BFGS is the recommended gradient-based alternative, achieving quasi-Newton convergence speed with only gradient evaluations and Wolfe-condition step control.
The five cases expose a clear progression:

1. The linear case establishes the baseline: conjugate Gibbs reference, mild variance collapse, all methods converge cleanly.

2. The quadratic case raises the parameter count and reveals that moderate variance collapse can persist even when the ELBO has converged.

3. The logistic case introduces a non-conjugate computational bottleneck, resolved by the Jaakkola–Jordan bound, and demonstrates that variance collapse can be avoided entirely when the bound is tight at the CAVI fixed point.

4. The hierarchical linear case adds group random effects ($J = 5$, $D = 19$), demonstrating that the same mean-field VI framework scales to mixed-effects models with minimal addi-

tional modelling effort. Asymmetric variance collapse emerges from the entanglement of β_0 with the random effects.

5. The hierarchical logistic case combines the JJ bound with random effects ($J = 5$, $D = 17$), completing the progression. Variance collapse becomes symmetric across both coefficients, and PG Gibbs confirms the VI solution.

The Pólya–Gamma Gibbs sampler provides an accurate reference for the logistic and hierarchical logistic cases but is computationally expensive ($n \times T$ Gamma draws per iteration). For production settings with $n \gg 200$, the VI methods offer a substantial computational advantage — CAVI is 1050× faster than PG Gibbs in Case Study 5.

References

Jaakkola, T. S., & Jordan, M. I. (2000). Bayesian parameter estimation via variational methods. *Statistics and Computing*, *10*(1), 25–37. https://doi.org/10.1023/A:1008932416310

Polson, N. G., Scott, J. G., & Windle, J. (2013). Bayesian inference for logistic models using Pólya–Gamma latent variables. *Journal of the American Statistical Association*, *108*(504), 1339–1349. https://doi.org/10.1080/01621459.2013.829001

A Case Study Summary Table

The following table expands on Table 1 with additional optimisation-specific and sampler-specific properties.

Table 2: Extended case study comparison.

Property	Linear	Quadratic	Logistic	Hier. Lin.	Hier. Log.
Model					
Likelihood	Gaussian	Gaussian	Bernoulli	Gaussian	Bernoulli
True $\boldsymbol{\beta}$	$(1.0, 2.0)$	$(1.0, 2.0, -1.0)$	$(-1.0, 2.0)$	$(1.0, 2.0)$	$(-1.0, 2.0)$
Predictors p	2	3	2	2	2
Observations n	50	100	200	150 ($J = 5$)	150 ($J = 5$)
Conjugacy	Yes	Yes	No	Yes	No
Variational approximation					
Variational family	Normal–Gamma	Normal–Gamma	Gaussian	Normal–Gamma	Gaussian
ELBO form	Closed form	Closed form	JJ bound	Closed form	JJ + hier.
VI parameters D	8	11	5	19	17
Variance collapse	Mild	Moderate	Minimal	Asymmetric	Symmetric
Reference sampler					
Sampler type	Conj. Gibbs	Conj. Gibbs	PG Gibbs	Hier. Gibbs	PG Gibbs
Cost per sweep	$O(p^3)$	$O(p^3)$	$O(n \cdot T)$	$O(N + J)$	$O(n \cdot T)$
ESS (typical)	High	High	≈ 250–280	High	Moderate
Optimisation					
CAVI convergence	< 50 iter	< 50 iter	< 50 iter	≈ 100 iter	≈ 37 iter
BFGS convergence	Fast	Fast	Fast	Fast	Fast
Recommended method	CAVI	CAVI	CAVI	CAVI	CAVI

The text associated with the logistic case for presentation

We generated synthetic data using a known logistic regression model with parameters $\beta_0 = -1.0$ and $\beta_1 = 2.0$. These are the data-generating parameters: they define the process that produced the observations and represent what we are trying to recover.
We then treated inference as if the true values were unknown. The Gibbs sampler uses only two inputs: the 200 binary observations and a weakly informative prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\boldsymbol{I})$. It never sees the true values.
The posterior distribution is the result of combining what the data say (the likelihood) with what we believed before seeing the data (the prior). The posterior mean sits at approximately $(-1.6, 2.6)$, offset from the true values, because with only 200 observations the likelihood is not strong enough to overcome the prior completely.
The true values appear on the plots only as a benchmark — to verify that the posterior is plausible. The fact that the true values fall within the posterior distribution confirms the sampler is behaving correctly.

B Individual Report References

The five individual case study reports contain full derivations, all figures, and complete code listings.

CS1 — Linear regression: `report/20260510A-LINEAR-CASE-STUDY.tex`. Full ELBO derivation, gradient verification, ELBO landscape, all convergence figures, posterior comparison figures, and variance collapse analysis.

CS2 — Quadratic regression: `report/20260510A-QUADRATIC-CASE-STUDY.tex`. Identical framework to the linear case with $p = 3$ and feature expansion.

CS3 — Logistic regression: `report/20260510A-LOGISTIC-CASE-STUDY.tex`. Full Jaakkola–Jordan derivation, Pólya–Gamma sampler, ESS diagnostics, and all 20 result figures.

CS4 — Hierarchical linear regression: `report/20260510A-HIERARCHICAL-LINEAR-CASE-STUDY.tex`. Two-level Normal–Gamma ELBO, group random effects, three-chain Gibbs sampler, and all 23 result figures. Asymmetric β_0 variance collapse.

CS5 — Hierarchical logistic regression: `report/20260510A-HIERARCHICAL-LOGISTIC-CASE-STUDY.tex` JJ bound with random effects, Pólya–Gamma Gibbs reference, and all 20 result figures. Symmetric variance collapse across both coefficients.

C Code Structure

All code is in the `python/` directory. Each notebook follows a five-stage structure.

Stage 1: Data generation. Saves `results/data/<case>_data.parquet`.

Stage 2: ELBO evaluation and gradient verification (finite-difference check).

Stage 3: Optimisation (CAVI, gradient ascent, Newton, BFGS). Saves `results/data/<case>_opt.parquet`.

Stage 4: Reference Gibbs sampler with ESS diagnostics. Saves `results/data/<case>_gibbs.parquet`.

Stage 5: Diagnostics and comparison figures. Saves `results/tables/<case>_diagnostics.csv`.

Case	Notebook	Figure prefix
Linear	`python/linear_run.ipynb`	`linear_`
Quadratic	`python/quadratic_run.ipynb`	`quadratic_`
Logistic	`python/logistic_run.ipynb`	`logistic_`
Hier. Linear	`python/hierarchical_linear_run.ipynb`	`hierarchical_linear_`
Hier. Logistic	`python/hierarchical_logistic_run.ipynb`	`hierarchical_logistic_`

Core algorithms are in `python/vb_algorithms_py.py`; utilities (archive, plot, I/O) are in `python/vb_utils_py.py`.
All five individual case study reports are available online at:

https://ENEL445.IndustrialOrigami.ai